

```
import numpy as np
a = np.array([1, 2, 3, 4, 5])
b = np.arange(6, 11)
c = np.linspace(0, 1, 5)
d = a * 2
e = np.mean(a)
f = np.median(a)
g = np.std(a)
print(a, b, c, d, e, f, g)
```

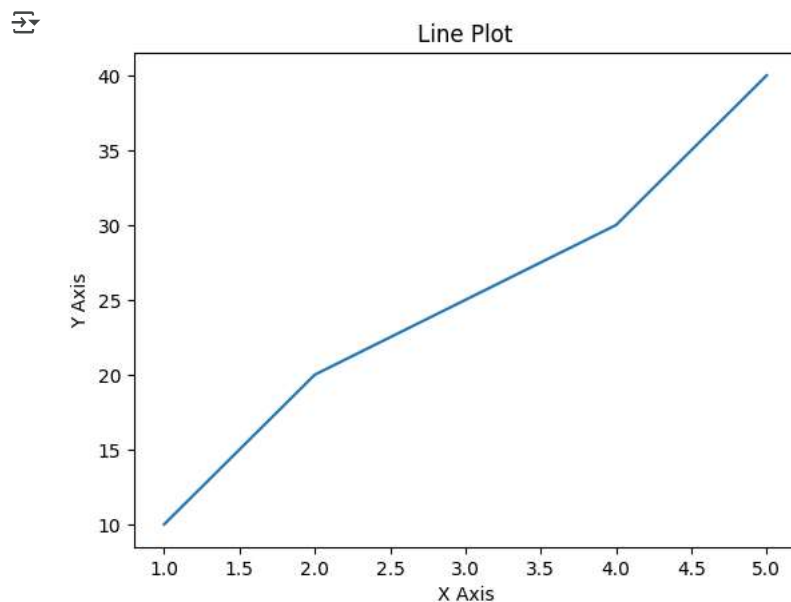
```
[1 2 3 4 5] [ 6  7  8  9 10] [0.  0.25 0.5  0.75 1.  ] [ 2  4  6  8 10] 3.0 3.0 1.4142135623730951
```

```
import pandas as pd
data = {'Name': ['A', 'B', 'C'], 'Score': [85, 90, 78]}
df = pd.DataFrame(data)
mean_score = df['Score'].mean()
max_score = df['Score'].max()
print(df)
print(mean_score, max_score)
```

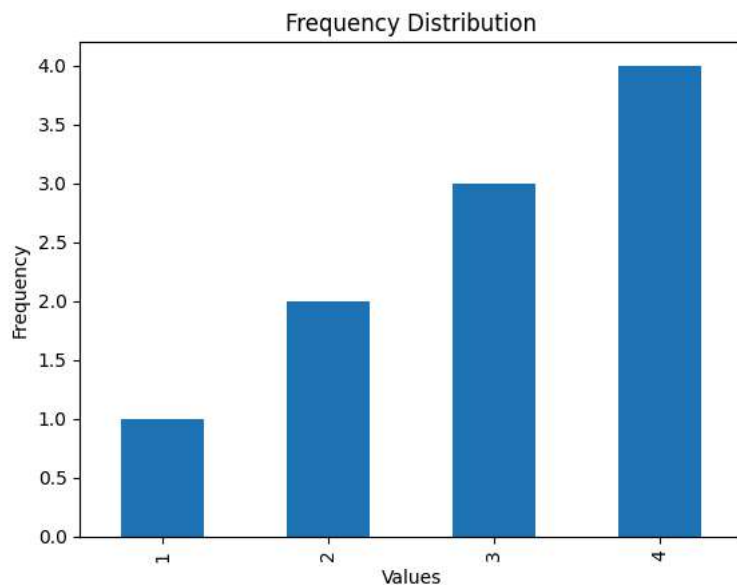
```

  Name  Score
0    A     85
1    B     90
2    C     78
84.33333333333333 90
```

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]
plt.plot(x, y)
plt.title("Line Plot")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```



```
import pandas as pd
import matplotlib.pyplot as plt
data = [1, 2, 2, 3, 3, 3, 4, 4, 4, 4]
df = pd.DataFrame(data, columns=['Values'])
freq = df['Values'].value_counts().sort_index()
freq.plot(kind='bar')
plt.title("Frequency Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```



```
import numpy as np
data = np.array([5, 10, 15, 20, 25])
variance = np.var(data)
std_dev = np.std(data)
range_val = np.ptp(data)
print(variance, std_dev, range_val)
```

50.0 7.0710678118654755 20

```
import numpy as np
data = np.array([10, 20, 30, 40, 50])
mean = np.mean(data)
median = np.median(data)
mode = np.bincount(data).argmax()
print(mean, median, mode)
```

30.0 30.0 10

```
import matplotlib.pyplot as plt
import numpy as np
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 4, 5])
plt.scatter(x, y)
plt.title("Scatter Plot")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```



Scatter Plot



```
import numpy as np
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 4, 5])
correlation = np.corrcoef(x, y)[0, 1]
print(correlation)
```



```
0.7745966692414834
```

```
3.0
```

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
x = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y = np.array([1, 3, 2, 3, 5])
model = LinearRegression().fit(x, y)
y_pred = model.predict(x)
plt.scatter(x, y)
plt.plot(x, y_pred, color='red')
plt.title("Linear Regression")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```



Linear Regression

